Statistics, Counting & Probability

Section 1: Expected Value & Probability Distributions

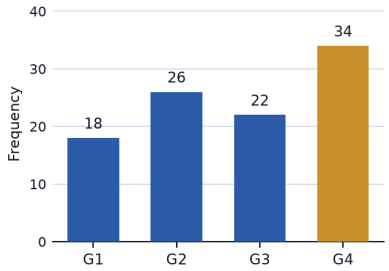
1 A charity raffle sells $1,000$ tickets at 5 each. There is one grand prize of $1,000$ and two second prizes of 500 each. What is the expected net gain for a person purchasing one ticket?

a. -$3.00

b. -$2.00

c. -$1.50

d. $0.00



2 A random variable X has probability distribution: $P(X=1)=0.2, P(X=2)=0.5, P(X=3)=0.3$. What is the expected value $E(X)$?

a. 2.0

b. 2.1

c. 2.2

d. 2.3

3 If $E(X)=4$ and $E(X^2)=25$, what is the variance $\text{Var}(X)$?

a. 3

b. 5

c. 9

d. 16

Section 2: Normal Distribution & Z-Scores

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4 The scores on a standardized exam are normally distributed with a mean of 500 and a standard deviation of 100. Approximately what percentage of test takers score between 400 and 600?

a. 50%

b. 68%

c. 95%

d. 99.7%

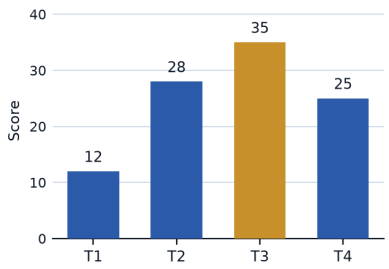
5 A student scores 84 on a test with mean 72 and standard deviation 8. What is the student's z-score?

a. 1.25

b. 1.50

c. 1.75

d. 2.00



6 In a normal distribution with mean 50 and standard deviation 5, what value corresponds to a z-score of -1.6?

a. 42

b. 44

c. 46

d. 48

Section 3: Conditional Probability & Bayes' Rule

7 Given $P(A) = 0.6, P(B) = 0.5,$ and $P(A \cap B) = 0.3$, what is the conditional probability $P(A \mid B)$?

a. 0.50

b. 0.60

c. 0.75

d. 0.30

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8 If two events A and B are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \cup B)$?

a. 0.70

b. 0.80

c. 0.90

d. 0.1575

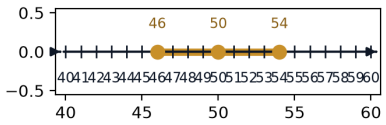
9 A medical screening test is 95% accurate. If 2% of a population has the disease, what is the probability that a randomly chosen person has the disease AND tests positive?

a. 0.019

b. 0.020

c. 0.095

d. 0.95



Section 4: Combinatorial Counting & Pigeonhole Principle

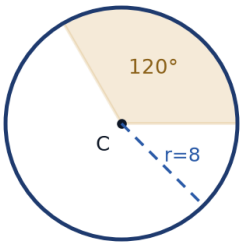
10 In how many ways can 6 students be seated in a circle?

a. 120

b. 720

c. 60

d. 24



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11 What is the minimum number of socks that must be drawn from a drawer containing 10 black socks and 10 white socks to guarantee getting at least one matching pair?

a. 2

b. 3

c. 4

d. 11

12 How many distinct permutations can be formed using all the letters of the word 'STATISTICS'?

a. $50,400$

b. $100,800$

c. $302,400$

d. $3,628,800$

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